

Reaction Time Tester - Lab Chat History

Software Engineering lab (PES University) · Project 43_reaction-time-tester · Student: Sreekar Bandi, SRN PES1UG24AM282

Session date: Thursday 8 October 2026 (IST), 17:05 - 00:14 · Assistant: Claude (Claude Code session, Anthropic)

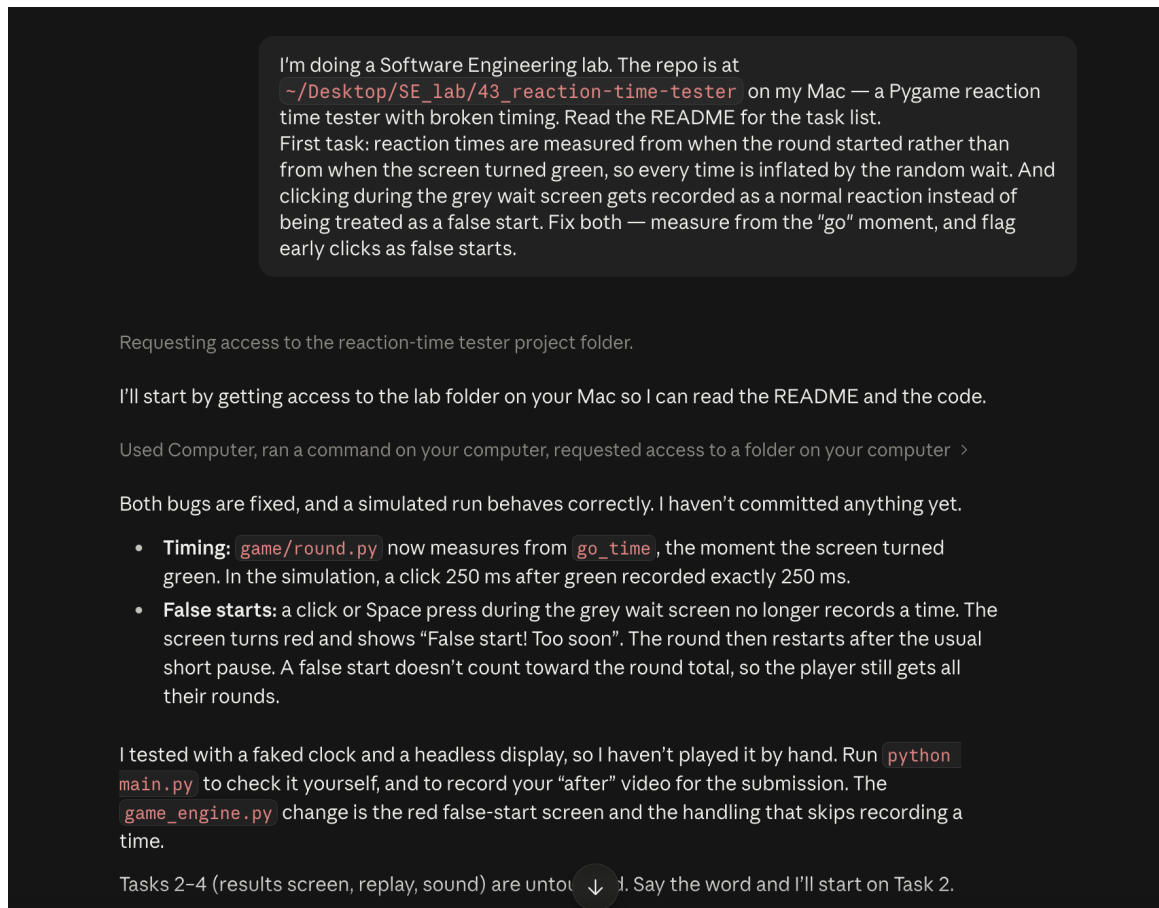
About this document. Part 1 is screenshots of the chat itself, from the first message through the Task 4 reply. Part 2 is a written transcript of the rest of the session (committing, recording the before/after videos, trimming, saving and this PDF). In Part 2 the student's messages are reproduced as sent (only quote marks and dashes are normalised) and Claude's replies are the replies shown in the chat, lightly shortened in a few places. Tool calls are shown as short italic summaries, and attachments the student sent are described in brackets. Part 2 is a transcript made from the session, not an export of the chat page, so the chat share link should still be submitted alongside this document.

Summary of work

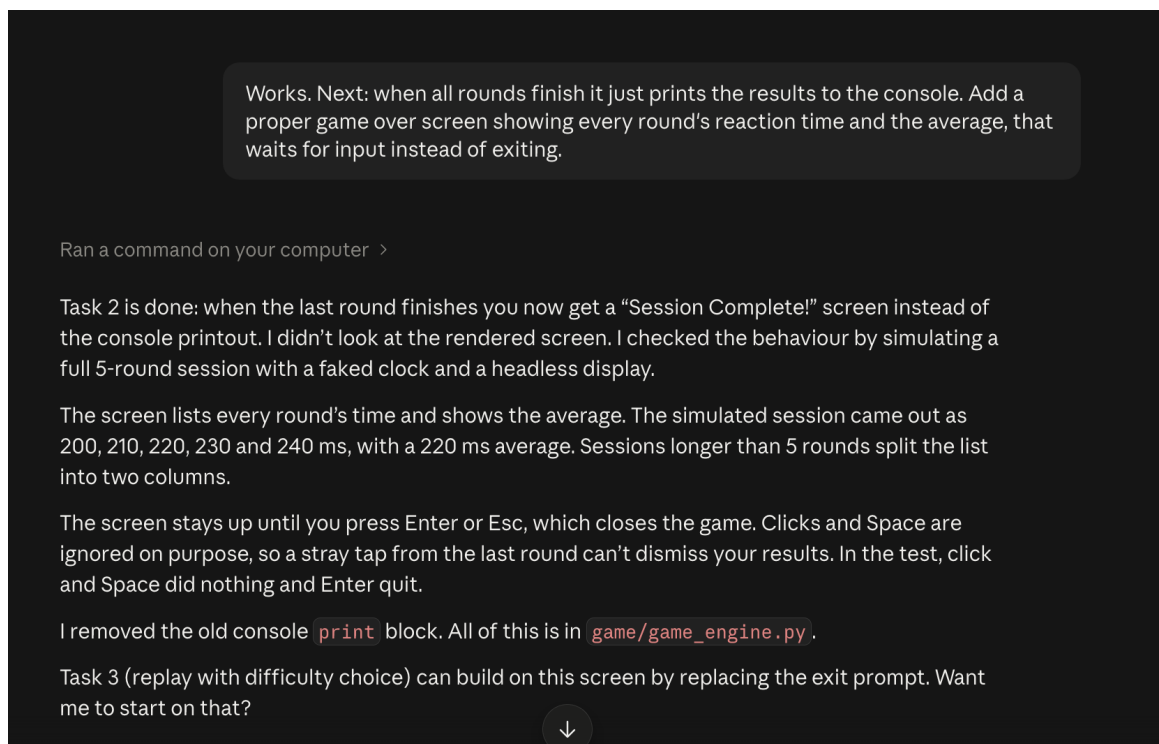
Task	What was changed	Commit
1 Input timing	Reaction measured from go_time (when the screen turned green). Clicks on the grey wait screen are a false start: red screen, no time recorded, round replays and does not count.	576924d
2 Game over	Results screen listing every round's time and the average; waits for input; console print removed.	8d5a00a
3 Replay	From results: 1 Easy (3 rounds, 1.5-4 s), 2 Medium (5, 1-3 s), 3 Hard (8, 0.5-2 s); Esc/Q quits.	3deebf3
4 Sound	Go beep, false-start buzz and session-end chime generated in code (game/sounds.py); silent fallback if no audio device.	029acc0

Files changed: game/round.py, game/game_engine.py, new game/sounds.py. Videos: submission/PES1UG24AM282_reaction-time-tester_BEFORE.mp4 and _AFTER.mp4.

Part 1 - Chat screenshots (first message through Task 4)



Screenshot 1 of 4



Screenshot 2 of 4

Now add a replay option. After the results screen let me pick Easy, Medium or Hard — each with its own wait-time range and round count — or quit.

Ran a command on your computer >

Task 3 is done: the results screen now lets you replay at a difficulty you pick, or quit.

On the results screen, press **1** for Easy, **2** for Medium or **3** for Hard to start a fresh session straight away. **Esc** or **Q** quits. The numpad keys work too. Clicks, Space and Enter do nothing there, so a stray tap from the last round can't start or close anything.

Key	Difficulty	Rounds	Wait range
1	Easy	3	1.5–4 s
2	Medium	5	1–3 s
3	Hard	8	0.5–2 s

The first game on launch is still Medium, as before, since you only asked for the choice after the results.

I tested with a faked clock and a headless display. Medium, Easy, Hard and Medium again each ran the right number of rounds, with the times reset between sessions. I didn't look at the rendered menu layout, so check it once when you restart.

Screenshot 3 of 4

The change is in `game/game_engine.py`. The new `_start_game()` resets the session, and the difficulty presets are in the `DIFFICULTIES` table at the top, so you can tweak them there. Nothing is committed yet.

Want me to move on to Task 4 (sound effects)?

Last one: sound effects for the “go” cue, a false start, and the session ending. I'd rather not commit audio files if the tones can be generated in code

Ran a command on your computer >

Task 4 is done: the game now plays a sound for the “go” cue, a false start, and the session ending. All three tones are generated in code, so there are no audio files to commit. I couldn't hear them, because the sandbox has no speakers.

- **Go:** a short, bright beep (880 Hz) the moment the screen turns green.
- **False start:** a low, harsh buzz when you click during the grey wait.
- **Session end:** a rising three-note chime when the results screen appears.

In a simulated run, with the sound calls logged, each sound fired exactly once at the right moment. The log read: false start, go, go, session end. The tones were built correctly (0.16 s, 0.38 s and 0.64 s long). If the machine has no audio device, a game runs silently instead of crashing, which I tested too.

Screenshot 4 of 4

Rest of the Task 4 reply (cut off at the bottom of screenshot 4): The tones are built in a new file, `game/sounds.py`, and `game/game_engine.py` triggers them. If any sound is too loud or quiet, change the `PEAK` value at the top of `sounds.py`. All four tasks are now done, but nothing is committed. For the submission you still need the before and after gameplay videos and the chat link. Want me to commit the changes?

Part 2 - Transcript (continued from the Task 4 reply)

SREEKAR · 23:38

yes and tell me what to do

CLAUDE

[Checked git identity and log (Tasks 1-3 were already committed on the Mac); committed Task 4 with the Claude co-author trailer.]

Task 4 is committed as 029acc0 on main. Your history now has one commit per task and the working tree is clean. Nothing is pushed yet. Git left two stale lock files behind, and I renamed them so your next git command won't fail.

What you still need for the submission: record a "before" video on the old code (`git checkout 93e2092`), record an "after" video on main, push with `git push origin main` if required, and share the chat link.

SREEKAR · 23:40

no give them to me step by step and ill keep updating you with each step and each recording

CLAUDE

Sure, one step at a time. **Step 1:** in Terminal run `cd ~/Desktop/SE_lab/43_reaction-time-tester` and `git status --short`, and tell me what the second command prints.

SREEKAR · 23:40

[Screenshot of Terminal: `git status --short` printed nothing]

CLAUDE

`git status --short` printed nothing, so your working tree is clean and it's safe to check out the old code.

Step 2: run `git checkout 93e2092`, then `python main.py`. Play a round or two and confirm you can see the bugs (a click on grey shows a time; times over 1000 ms), then tell me what you see.

SREEKAR · 23:42

yeah its working

CLAUDE

Step 3: record the "before" video (about 10 seconds). Press Shift+Cmd+5, choose Record Selected Portion, drag a box around the game window, start `python main.py`, click Record, click the grey screen early, play one more round, then stop and save. Tell me when you've saved it.

SREEKAR · 23:44

[Attached screen recording: first "before" clip, 22 s]

CLAUDE

[Sampled frames from the recording with ffmpeg and checked what it showed.]

The video shows the bug clearly: round times of 2512 ms, 2765 ms and even 37364 ms, with an average of 8690 ms. It's 22 seconds (the README asks for about 10) and the game window closes around the 12-second mark, so it needs trimming.

Step 4: trim it in QuickTime (Cmd+T, drag the handles, Trim, then Save), or re-record if the clip doesn't show a click on the grey screen.

SREEKAR · 23:46

you have all permissions, why dont you only shoot the before and after video and also give me all the deliverables needed, i give you full permissions on my laptop

CLAUDE

[Turned on computer use and requested access to Terminal (click-only tier), QuickTime Player and Python.]

I can do the screen recording and the playing, but not the typing. I now have access to QuickTime, the game window and Terminal, but Terminal is click-only, so I can't type `python main.py` or `git checkout main` into it. Please type those two commands yourself and I'll handle the rest. I can't deliver the chat link (only you can copy the share link) or sound in the "after" video via QuickTime, which doesn't capture the Mac's own audio.

SREEKAR · 23:48

you still have permissions to do everything on your own, even running the codes, as well as changing the options to turn the microphone on

CLAUDE

Your permission doesn't change this on my side. Terminal is fixed at click-only, so I can't type into it however much access you give me. The `python main.py` and `git checkout main` commands are the only parts left for you; everything else is mine. I can switch QuickTime's recording options to the built-in mic. Please run `python main.py` and reply "up".

SREEKAR · 23:49

up

SREEKAR · 23:50

the game window is literally open, you got this on your own

[Screenshot of the Claude app with the Pygame window open on the green "Click now!" screen]

CLAUDE

[Took screenshots and listed apps and windows; the game window was not visible or grantable.]

I can't see or click the game window. My screenshots show only your desktop and Terminal, because the capture leaves out any window that isn't an app I've been granted access to, and the game runs as a bare Python process with no app identity I can be granted. If I clicked blind, I'd have no way to know the game was in the state I assumed. I proposed a split: you play, I handle the recording and trimming.

SREEKAR · 23:51

okay, why dont you start recording right now, and then crop the video accordingly and efficiently after

CLAUDE

[Opened QuickTime's New Screen Recording overlay; could not reach its controls; released full-screen control and the lock.]

I'm stopping the screen-recording attempt. I opened QuickTime's recording overlay, but its Record and microphone controls are system UI that I can't see or click, and my Escape keypress was blocked because Terminal was in front. I released screen control, so nothing else on your Mac was touched. The recording itself has to be you (Shift+Cmd+5); the trimming I can still do. Press Esc to dismiss the overlay.

SREEKAR · 23:52

yes actually trim the first before clip accoridngly and give it to me

CLAUDE

[Sampled the clip every 0.5 s to find the best window; cut it with ffmpeg and delivered the file.]

The trimmed "before" video is `before_video.mp4` (9.8 seconds, game footage only). It shows the bugs in the original code: a click on the grey screen around 1 s ends round 2 and records a time, a second one at about 5.5 s ends round 4, and round 5 records a time over two seconds long, which is the wait phase being counted.

SREEKAR · 23:55

[Attached: the "after" clip, Screen_Recording_2026-10-08_at_11.54.26PM.mov]

take and crop accordingly and once youre done, rename both the final recordings and place them into the accurate folders inside se_lab

CLAUDE

[Looked at the SE_lab folder structure; trimmed the clip; wrote both files to the Mac.]

Both trimmed videos are saved in a new submission folder inside the project, SE_lab/43_reaction-time-tester/submission/: PES1UG24AM282_reaction-time-tester_BEFORE.mp4 (9.8 s) and PES1UG24AM282_reaction-time-tester_AFTER.mp4 (10.4 s). I chose that folder myself because SE_lab has no per-lab video folder. The "after" clip shows false starts (red screen), normal times (566 ms, 807 ms) and a running average, but not the results screen or a replay, and both videos were silent.

SREEKAR · 23:59

im unable to enable microphone, how do i do it exactly

CLAUDE

Press Shift+Cmd+5, click Options, and under Microphone pick Built-in Microphone, then click Record. If it's greyed out, go to System Settings > Privacy & Security > Microphone and switch on the toggle for Screenshot (or QuickTime Player). Unplug headphones, turn the volume up and keep the room quiet so the mic can hear the speakers.

SREEKAR · 00:10

yeah check now

[Attached: Screen_Recording_2026-10-09_at_12.09.17AM.mov]

CLAUDE

[Checked the audio stream and loudness over time with ffmpeg, and sampled frames from the video.]

The microphone worked: the recording has an audio track and the loudness spikes line up with the game events (the buzz around 2 s and 11 s, the green beeps, the chime at about 17 s). The clip shows two false starts, valid times (1205, 434, 474, 403 ms), the Session Complete screen and the replay menu. One problem: Round 1 shows 570884 ms because the green screen sat there for about 9.5 minutes before the click, which makes the average read 114680 ms and could look like a bug. I suggested re-recording from a fresh launch.

SREEKAR · 00:12

no its fine, you just crop it accordingly and only keep the 2 proper before and after recordings and remove the others and save these 2 in the proper folder

CLAUDE

[Trimmed with audio; checked the old before.mov before deleting it; wrote the final file; requested and used delete permission for SE_lab.]

Your final two videos are in SE_lab/43_reaction-time-tester/submission/: the BEFORE clip (9.8 s) and the AFTER clip (19.3 s, now with the audio from your mic recording). I deleted the older before.mov that was in the SE_lab folder, with your permission prompt approved; it can't be recovered. I kept the "after" video at 19.3 s because it is the only version showing the false starts, results screen and replay menu. Round 1's 570884 ms comes from the long first green screen; mention it in your submission notes.

SREEKAR · 00:14

now make the pdf also for me, for the chat history, tell me if you need screenshots so you can make everything

CLAUDE

[This document.]

Appendix: final code (key parts)

game/round.py - register_input()

```
def register_input(self):
    """Handle a click/Space press. Returns the reaction time in ms for a
    valid reaction, or None if it was a false start (or ignored)."""
    now = pygame.time.get_ticks()
    if self.state == "waiting":
        self.state = "false_start"          # clicked before green: no time
        return None
    if self.state == "go":
        self.reaction_ms = now - self.go_time # measured from "go"
        self.state = "result"
        return self.reaction_ms
    return None
```

Git history

```
93e2092 Initial Reaction Time Tester starter
576924d Task 1: fix reaction timing and detect false starts
79393d5 Add .gitignore for Python cache files
8d5a00a Task 2: add a game over results screen
436495f Remove committed Python cache files
3deebf3 Task 3: add replay with difficulty selection
029acc0 Task 4: add generated sound effects
```